

2.11 Exercises

2.11.1 Problems

2.1 Consider the following five observations. You are to do all the parts of this exercise using only a calculator.

$$\hat{x} = 1$$

$$\hat{y} = 2$$

x	y	$x - \bar{x}$	$(x - \bar{x})^2$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$
3	4	2	4	2	4
2	2	1	1	0	0
1	3	0	0	1	0
-1	1	-2	4	-1	2
0	0	-1	1	-2	2
$\sum x_i = 5$	$\sum y_i = 10$	$\sum (x_i - \bar{x}) = 0$	$\sum (x_i - \bar{x})^2 = 10$	$\sum (y_i - \bar{y}) = 0$	$\sum (x_i - \bar{x})(y_i - \bar{y}) = 8$

- Complete the entries in the table. Put the sums in the last row. What are the sample means $\bar{x}$ and $\bar{y}$?
- Calculate b_1 and b_2 using (2.7) and (2.8) and state their interpretation.
- Compute $\sum_{i=1}^5 x_i^2$, $\sum_{i=1}^5 x_i y_i$. Using these numerical values, show that $\sum (x_i - \bar{x})^2 = \sum x_i^2 - N\bar{x}^2$ and $\sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - N\bar{x}\bar{y}$.

a.

$$\sum x_i = 3 + 2 + 1 - 1 + 0 = 5 \Rightarrow \hat{x} = \frac{5}{5} = 1$$

$$\sum y_i = 4 + 2 + 3 + 1 + 0 = 10 \Rightarrow \hat{y} = \frac{10}{5} = 2$$

b.

$$\text{slop} = b_2 = \frac{\sum (x_i - \hat{x})(y_i - \hat{y})}{\sum (x_i - \hat{x})^2}$$

$$\frac{4 + 0 + 0 + 2 + 2}{4 + 1 + 0 + 4 + 1} = \frac{8}{10} = 0.8$$

x 增加 1 單位, y 增加 0.8

$$\text{intercept} = b_1 = \hat{y} - b_2 \hat{x}$$

$$2 - (0.8 \times 1) = 1.2$$

$x=0$ 時, $y=1.2$

$$\begin{aligned}
 C. \sum_{i=1}^5 x_i^2 &= 3^2 + 2^2 + 1^2 + (-1)^2 + 0^2 \\
 &= 9 + 4 + 1 + 1 + 0 \\
 &= 15
 \end{aligned}$$

$$\sum (x_i - \bar{x})^2 = \sum x_i^2 - N \hat{x}^2$$

$$\Rightarrow \text{column 3} = 15 - 5 \times (1)^2 = 10$$

$$\begin{aligned}
 \sum_{i=1}^5 x_i y_i &= 3 \times 4 + 2 \times 2 + 1 \times 3 + (-1) \times 1 + (0 \times 0) \\
 &= 12 + 4 + 3 - 1 + 0 = 18
 \end{aligned}$$

$$\sum (x_i - \hat{x})(y_i - \hat{y}) = \sum x_i y_i - N \hat{x} \hat{y}$$

$$\Rightarrow \text{column 5}$$

$$= \underline{3 \times 4 + 2 \times 2 + 1 \times 3 + (-1) \times 1 + 0 \times 0} - \underline{5 \times (1) \times (2)}$$

$$= 18 - 10$$

$$= 8$$

$$\hat{y}_i = 1.2 + 0.8x_i$$

- d. Use the least squares estimates from part (b) to compute the fitted values of y , and complete the remainder of the table below. Put the sums in the last row.

Calculate the sample variance of y , $s_y^2 = \sum_{i=1}^N (y_i - \bar{y})^2 / (N - 1)$, the sample variance of x , $s_x^2 = \sum_{i=1}^N (x_i - \bar{x})^2 / (N - 1)$, the sample covariance between x and y , $s_{xy} = \sum_{i=1}^N (y_i - \bar{y})(x_i - \bar{x}) / (N - 1)$, the sample correlation between x and y , $r_{xy} = s_{xy} / (s_x s_y)$ and the coefficient of variation of x , $CV_x = 100(s_x / \bar{x})$. What is the median, 50th percentile, of x ?

x_i	y_i	$\hat{y}_i$	$\hat{e}_i$	$\hat{e}_i^2$	$x_i \hat{e}_i$
3	4	3.6	0.4	0.16	1.2
2	2	2.8	-0.8	0.64	-1.6
1	3	2.0	1.0	1	1.0
-1	1	0.4	0.6	0.36	-0.6
0	0	1.2	-1.2	1.44	0
$\sum x_i =$	$\sum y_i =$	$\sum \hat{y}_i =$	$\sum \hat{e}_i =$	$\sum \hat{e}_i^2 =$	$\sum x_i \hat{e}_i =$
5	10	10	0	3.6	0

- e. On graph paper, plot the data points and sketch the fitted regression line $\hat{y}_i = b_1 + b_2 x_i$.
- f. On the sketch in part (e), locate the point of the means $(\bar{x}, \bar{y})$. Does your fitted line pass through that point? If not, go back to the drawing board, literally.
- g. Show that for these numerical values $\bar{y} = b_1 + b_2 \bar{x}$.
- h. Show that for these numerical values $\bar{\hat{y}} = \bar{y}$, where $\bar{\hat{y}} = \sum \hat{y}_i / N$.
- i. Compute $\hat{\sigma}^2$.
- j. Compute $\widehat{\text{var}}(b_2 | \mathbf{x})$ and $\text{se}(b_2)$.

$$\textcircled{1} \dots \hat{e}_i = y_i - \hat{y}_i$$

d sample variance

樣本變異數 S_y^2 : $\frac{4+0+1+1+4}{5-1} = 2.5$

樣本變異數 S_x^2 : $\frac{4+1+0+4+1}{5-1} = 2.5$

sample covariance

樣本共變異數 S_{xy} : $\frac{4+0+0+2+2}{5-1} = 2$

sample correlation

相關係數 r_{xy} : $\frac{2}{(2.5)(2.5)} = 0.8$

coefficient of variation

變異係數 CV_x : $100\left(\frac{\sqrt{2.5}}{1}\right) \approx 158.11\%$

50th percentile of x

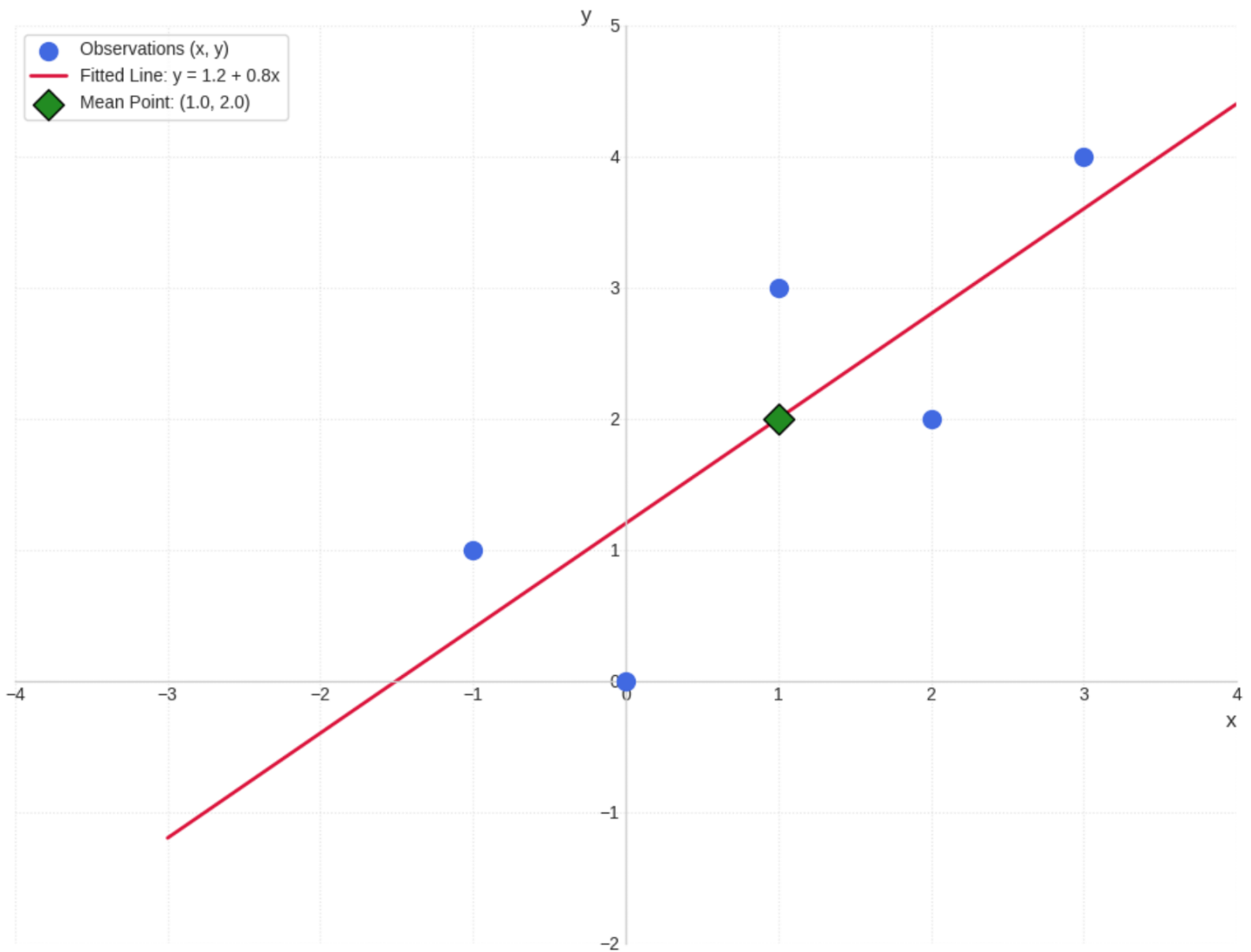
x的中位數 M_x : $\{-1, 0, 1, 2, 3\}$

$M_x = 1$

e & f

OLS regression line must pass through the sample mean points $(\hat{x}, \hat{y})$

Simple Linear Regression: Visualizing the Fitted Line and Means



g.

$$\text{intercept} = b_1 = \hat{y} - b_2 \bar{x}$$

$$\Rightarrow \hat{y} = b_1 + b_2 \hat{x}$$

h.

$$\hat{\bar{y}} = \frac{\sum (b_1 + b_2 x_i)}{N} = \frac{Nb_1 + b_2 \sum x_i}{N}$$

$$= b_1 + b_2 \hat{\bar{x}} = \hat{\bar{y}}$$

預測值平均 = 觀測值平均

i

誤差變異數估計值 $\hat{\sigma}^2$

$$\hat{\sigma}^2 = \frac{\sum \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$$

j

b_2 變異數估計值 $\widehat{\text{Var}}(b_2/x)$

$$\widehat{\text{Var}}(b_2) = \frac{\hat{\sigma}^2}{\sum (x_i - \bar{x})^2} = \frac{1.2}{10} = 0.12$$

b_2 標準誤 $se(b_2)$

$$se(b_2) = \sqrt{0.12} \approx 0.3464$$